

LESSBURG, VA, USA

WMO: 724055

Lat: 39.078N

Lon: 77.558W

Elev: 119

StdP: 99.91

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 03714

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.7	-19.6	0.7	-5.6	-17.0	0.9	-5.4	12.6	0.5	11.3	0.4	3.2	320	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	35.0	24.8	33.0	24.0	32.3	23.6	26.3	32.3	25.6	31.2	24.9	30.1	3.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.3	29.1	23.9	19.1	28.0	23.0	17.9	27.1	82.4	32.0	79.1	30.6	76.2	30.5	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.5	7.4	-13.8	37.1	2.9	1.2	-15.9	37.9	-17.6	38.6	-19.2	39.3	-21.3	40.1		
(5)			-14.4	27.8	2.7	0.6	-16.3	28.3	-17.9	28.6	-19.4	29.0	-21.3	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	1.5	2.9	7.4	13.3	18.4	23.3	25.6	24.8	20.8	14.3	8.1	3.6
	DBStd	9.58	5.56	5.29	5.25	4.90	4.31	3.42	2.77	2.80	3.75	4.49	4.70	4.73
	HDD10.0	921	270	206	116	21	1	0	0	0	0	10	92	207
	HDD18.3	2442	521	432	341	164	54	5	0	0	18	141	309	457
	CDD10.0	2282	7	7	35	120	261	398	483	460	325	144	33	8
	CDD18.3	761	0	0	2	14	57	153	224	202	93	16	1	0
	CDH23.3	7567	0	3	27	201	588	1488	2344	1970	787	151	8	1
	CDH26.7	3124	0	0	5	66	207	606	1073	853	277	36	1	0
Wind	WSAvg	2.5	3.0	3.1	3.3	3.0	2.3	2.1	1.9	1.7	2.0	2.3	2.6	2.7
Precipitation	PrecAvg	1033	72	59	87	88	102	94	96	86	102	83	74	81
	PrecMax	1556	175	145	185	200	223	212	206	159	238	185	172	143
	PrecMin	728	32	15	0	42	28	32	40	39	0	3	15	22
	PrecStd	187	35	31	38	38	48	47	39	29	64	52	35	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	22.3	27.1	32.4	33.0	35.9	37.2	36.4	34.0	30.1	24.0	21.1
		MCWB	15.8	15.4	17.1	19.7	22.9	25.2	25.1	25.3	23.2	22.3	16.8	16.6
	2%	DB	17.0	17.6	22.6	28.0	31.3	33.0	35.1	34.1	32.2	27.4	21.1	17.0
		MCWB	14.0	11.5	14.6	18.1	22.3	24.2	25.3	24.6	22.8	20.1	15.9	13.9
	5%	DB	12.9	13.9	19.7	25.3	28.9	32.2	33.0	32.6	30.0	24.8	18.0	13.0
		MCWB	10.0	8.9	13.2	17.0	21.0	23.7	24.3	23.9	22.1	18.9	14.1	9.8
	10%	DB	9.1	11.1	17.1	22.4	27.1	30.2	32.3	31.3	27.7	22.4	16.2	11.1
		MCWB	6.8	7.2	11.5	15.4	19.8	22.9	24.0	23.3	21.0	17.6	12.4	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	16.9	18.1	21.4	24.6	26.7	27.5	27.3	25.8	23.4	19.1	17.8
		MCDB	17.8	20.9	24.2	27.5	31.1	33.2	34.2	33.4	30.9	26.8	22.3	19.0
	2%	WB	14.3	13.0	16.4	19.8	23.4	25.5	26.5	26.2	24.6	22.1	17.4	14.8
		MCDB	16.8	16.3	21.1	26.1	29.2	31.5	32.6	31.9	29.0	25.9	19.1	17.0
	5%	WB	10.5	9.9	14.1	18.1	22.2	24.7	25.7	25.3	23.4	20.3	15.3	11.2
		MCDB	13.2	13.1	18.2	23.3	27.3	30.1	31.3	30.5	27.1	23.5	17.7	12.8
	10%	WB	7.2	7.6	11.9	16.6	20.9	23.9	25.0	24.5	22.4	18.2	13.0	8.0
		MCDB	8.8	9.6	16.2	21.4	24.9	28.6	30.2	29.3	25.9	21.2	15.6	9.9
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.8	11.1	12.4	11.7	11.3	11.5	11.4	11.3	11.4	10.5	8.7
		MCDBR	13.0	14.7	16.2	16.9	15.0	13.6	12.9	13.2	14.1	14.7	14.0	12.5
		MCWBR	10.2	10.2	9.6	8.5	7.1	5.6	5.0	5.0	5.9	8.0	9.5	9.7
	5% WB	MCDBR	12.0	13.1	13.9	14.0	13.3	11.9	11.5	11.7	11.1	11.8	11.3	11.2
		MCWBR	10.2	10.1	9.5	7.9	6.6	5.4	5.0	4.9	5.4	7.2	9.0	9.9
Clear-Sky Solar Irradiance	taub	0.305	0.321	0.356	0.400	0.440	0.504	0.518	0.518	0.432	0.362	0.322	0.307	
	taud	2.464	2.397	2.372	2.272	2.191	2.049	2.029	2.010	2.229	2.425	2.499	2.492	
	Ebn at Noon	871	901	900	876	845	789	772	760	810	842	844	841	
	Edh at Noon	80	98	111	131	145	167	169	168	127	94	77	73	
All-Sky Solar Radiation	RadAvg	2.02	2.87	3.82	4.83	5.43	6.00	5.88	5.25	4.30	3.21	2.31	1.77	
	RadStd	0.18	0.33	0.24	0.38	0.57	0.45	0.33	0.27	0.54	0.36	0.20	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	+0.38	+0.69	N/A	N/A	N/A	+77
(47)	Regional (65 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+125	+69

Nomenclature: See separate page